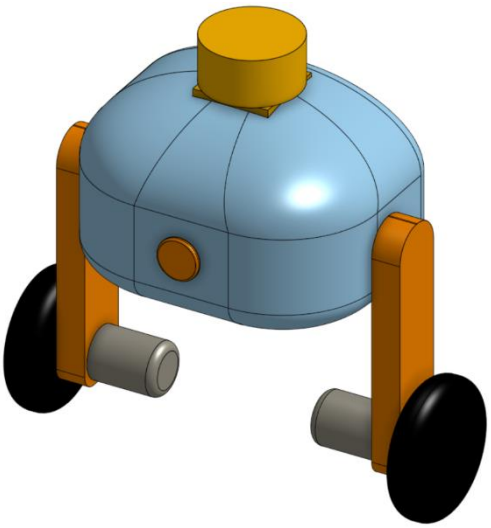
# Gyrobot use cases




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# Introduction

As part of our fourth year at Polytech Orléans engineering school, we are required to undertake one of the projects provided or to propose one of our own. We therefore proposed to build a robot that balances on two wheels using a gyroscope: the Gyrobot.

It is a compact, self-balancing, two-wheeled robot, designed to operate in environments where mobility, accessibility and autonomous navigation can pose challenges.

The project aims to explore the design of a small, lightweight robotic platform capable of maintaining its balance on two wheels, whilst serving as a basis for future applications such as mapping, exploration and assistance in challenging environments.

In this report, we will cover the general description of the robot, the user experience, the algorithmic operation, the use case, and finally the Gantt chart.

# 1. General description of the robot

The robot can be divided into three main parts:

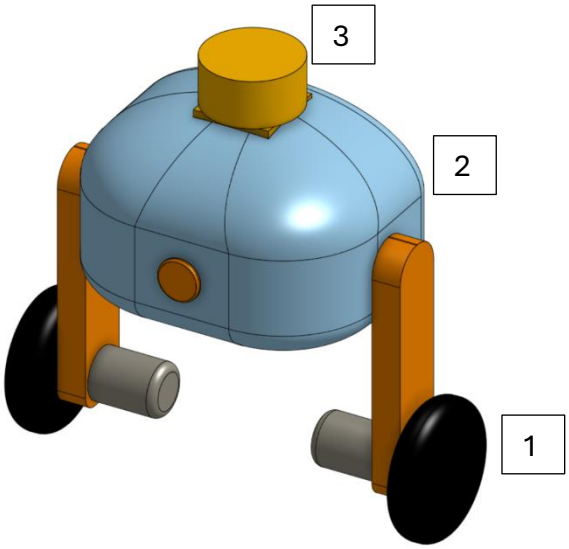



*Figure 1: 3D sketch of the Gyrobot*

The lower part of the robot, the motors, wheels and 'legs' marked **1** in the image below. This part must be robust so that it can withstand the rigours of various types of terrain as well as the impacts caused by the robot's movement.

The central section of the robot, labelled **2**, contains the various components required for movement, balance and the processing of lidar data. It also houses the battery needed to power the robot. The biggest challenge with this section will be managing the weight and its distribution.

The upper section of the robot, designated **3**, will house a tool required for mapping. This section must include at least two different tools in order to make the robot more versatile.

The project also includes a user interface to display the data transmitted and processed by the robot.

# 2. User experience

This section focuses on what the user needs to do to use our product.

## 2.1. Essential feature of the map

The Gyrobot will need to move around, map its surroundings, process the data and send it back to a user interface. It will also need to receive the commands required to move the robot. That is why we have opted to use two cards, to avoid a potential data overload if there were only one card.

A map will need to manage commands relating to the robot's movements. It must therefore:

- To be able to transmit the commands required for movement wirelessly
- Transmit data such as speed, the robot's angle relative to the ground and remaining battery charge
- Heat resistance
- To be fairly small in size
- To be energy-efficient

The second card will be responsible for processing mapping-related data and sending it to our HMI. It will therefore need to:

- To have sufficient computing power to process lidar data
- To be able to communicate wirelessly with the HMI
- Heat resistance
- To be fairly small in size
- To be energy-efficient

There are various boards to choose from: an Arduino, an ESP 32, a Raspberry Pi, an STM 32 and a Tensy.

The selection of the various components is carried out in: 7. Description of technical choices.

## 2.2. HMI

We also need to develop a user interface to display the data returned by the robot.

The image below was created partly with the help of generative AI.

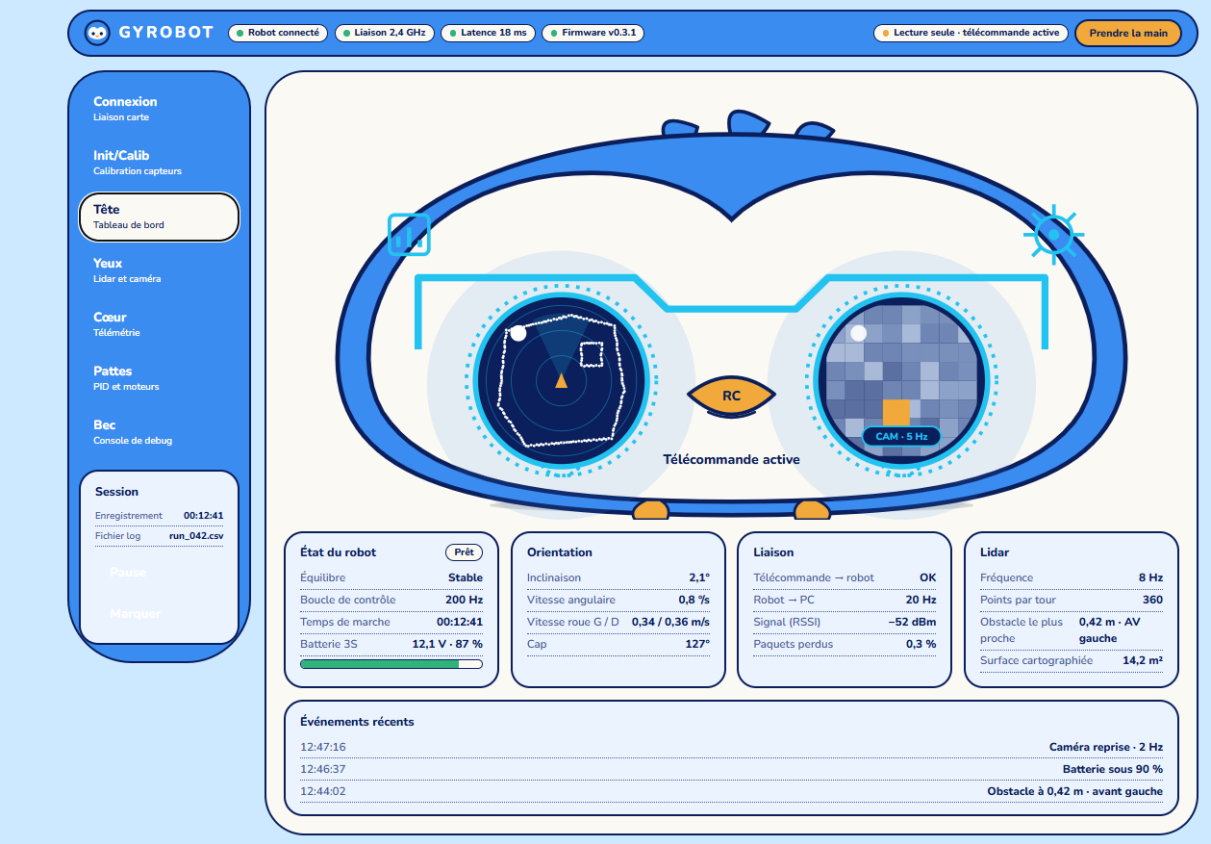



*Figure 2: Image from the IMH*

The user interface will comprise several sections/tabs to monitor the various values and lidar data returned by the robot. This penguin-themed user interface was chosen by our team and is purely for aesthetic purposes.

The layout and the data displayed in the user interface are subject to change.

# 3. Algorithmic operation

## 3.1. Flow chart

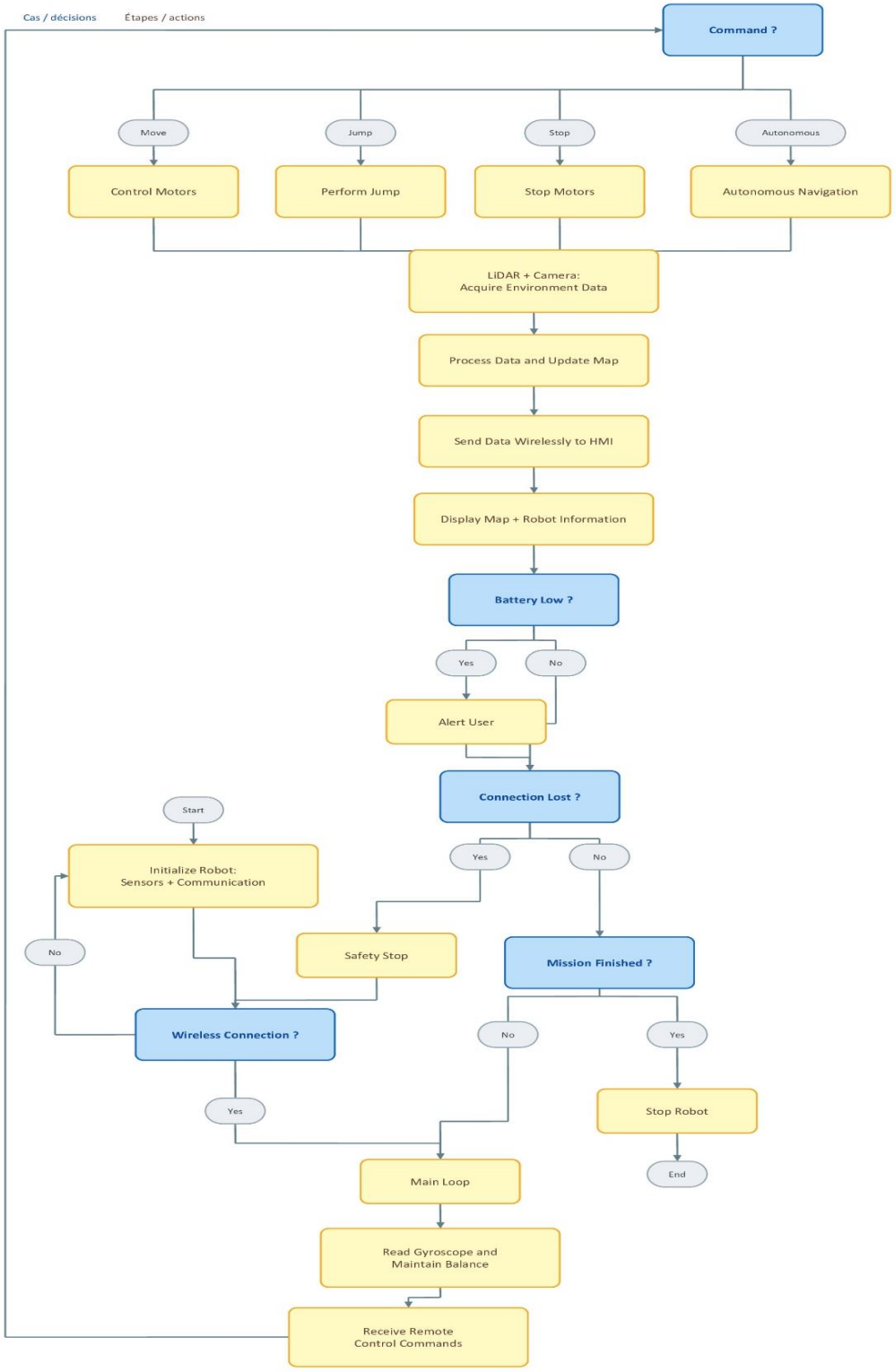



*Figure 3: Flowchart of the Gyrobot*

This flowchart explains how the programme's algorithm works.

# 4. Specifications

In order to draw up the specifications correctly, we need to identify the project's requirements and expectations. As regards expectations, we need to build a robot that can balance on two wheels, is capable of mapping its environment, and can interact via a remote control and a user interface that displays the robot's data. As for the requirements, we have defined two types: primary requirements, which are necessary for the robot's basic operation, and secondary requirements, which serve to make the robot more versatile.

The primary requirements align with our expectations, as we have set them ourselves.

As for the secondary requirements, we wish to develop a user interface and a robot resembling a penguin, to have a rechargeable battery, and for the lidar to be replaceable with a high-resolution camera.

# 5. Retro planning (Gantt)

**Gantt Gyrobot**

**T0 Lancement : 03/09/26 | T1 Audit : 06/10/26 | T2 PDR : 20/11/26 | T3 CDR + Mini salon : 08/01/27**

| Phase | Tâche | Responsable | Début | Fin | Durée (j) |
|---|---|---|---|---|---|
| | **T0 — Lancement du projet** | | 03/09/26 | 03/09/26 | 1 |
| Phase A | Mettre à jour GitLab | Tous | 03/09/26 | 05/10/26 | 33 |
| | Faire le ReadMe | Simon | 03/09/26 | 08/09/26 | 6 |
| | Faire les differents diagrammes | Tous | 04/09/26 | 22/09/26 | 19 |
| | Identifier les fonctions HVA du projet | Tous | 15/09/26 | 29/09/26 | 15 |
| | Cahier des charges | Issam | 21/09/26 | 30/09/26 | 10 |
| | Approche RSE | Maud | 30/09/26 | 04/10/26 | 5 |
| | Rédiger le rapport technique T1 (Audit) | Alexandre | 28/09/26 | 05/10/26 | 8 |
| | Faire le PPT (chemin critique) | Tous | 30/09/26 | 05/10/26 | 6 |
| | **T1 — Audit** | | 06/10/26 | 06/10/26 | 1 |
| Phase B — Conception & solutions | Détailler le Gantt de la phase T1 | À définir | 06/10/26 | 07/10/26 | 2 |
| | Choix des composants | À définir | 29/09/26 | 15/10/26 | 17 |
| | Générer et évaluer des solutions techniques pour chaque fonction | À définir | 06/10/26 | 19/10/26 | 14 |
| | Avancer la conception mécanique 3D du système (dimensionnement, architecture) | À définir | 06/10/26 | 03/11/26 | 22 |
| | Sélectionner et commander les composants nécessaires | À définir | 13/10/26 | 19/10/26 | 7 |
| | Justifier les solutions techniques retenues + volet RSE | À définir | 03/11/26 | 19/11/26 | 17 |
| | Revue préliminaire des HVA + mise à jour du rapport technique (PDR) | À définir | 02/11/26 | 20/11/26 | 19 |
| | **T2 — PDR** | | 20/11/26 | 20/11/26 | 1 |
| Phase C — Intégration & validation HVA | Détailler le Gantt de la phase T2 | À définir | 20/11/26 | 21/11/26 | 2 |
| | Fabriquer / assembler le système (mécanique + électronique) | À définir | 23/11/26 | 04/01/27 | 29 |
| | Développer et intégrer les fonctions HVA retenues | À définir | 24/11/26 | 05/01/27 | 29 |
| | Tester chaque fonction par rapport aux exigences (matrice de conformité) | À définir | 15/12/26 | 05/01/27 | 8 |
| | Résoudre les défauts identifiés | À définir | 04/01/27 | 07/01/27 | 4 |
| | Préparer la démonstration des HVA (CDR) + vidéo marketing (anglais) pour le mini salon | À définir | 15/12/26 | 07/01/27 | 10 |
| | Rédiger le rapport technique final S1 + auto-évaluation | À définir | 15/12/26 | 07/01/27 | 10 |
| | **T3 — CDR + Mini salon** | | 08/01/27 | 08/01/27 | 1 |

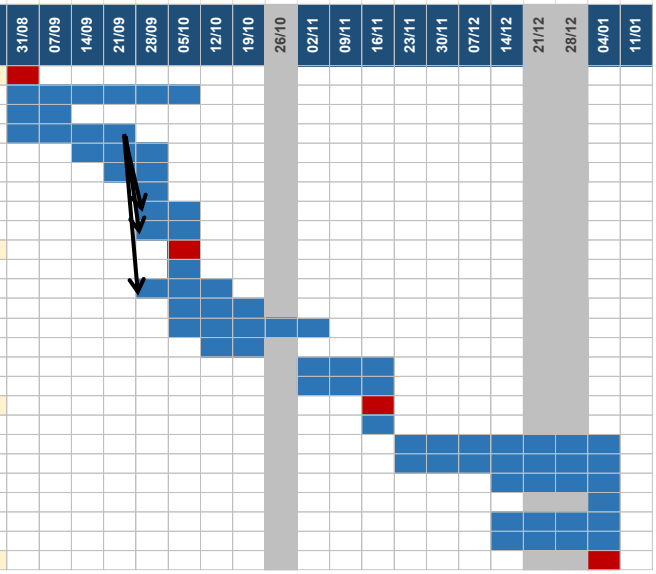



*Table 1: Gantt chart (updated on 21 September 2026)*

This schedule will be updated as we make progress on the project. In terms of organisation, our biggest challenge is that we only have three months to build a prototype. This means we need to have a very precise and rigorous organisational structure.

# 6. Validation testing and functional analysis

## 6.1. Functional analysis

### 6.1.1. Horned beast diagram

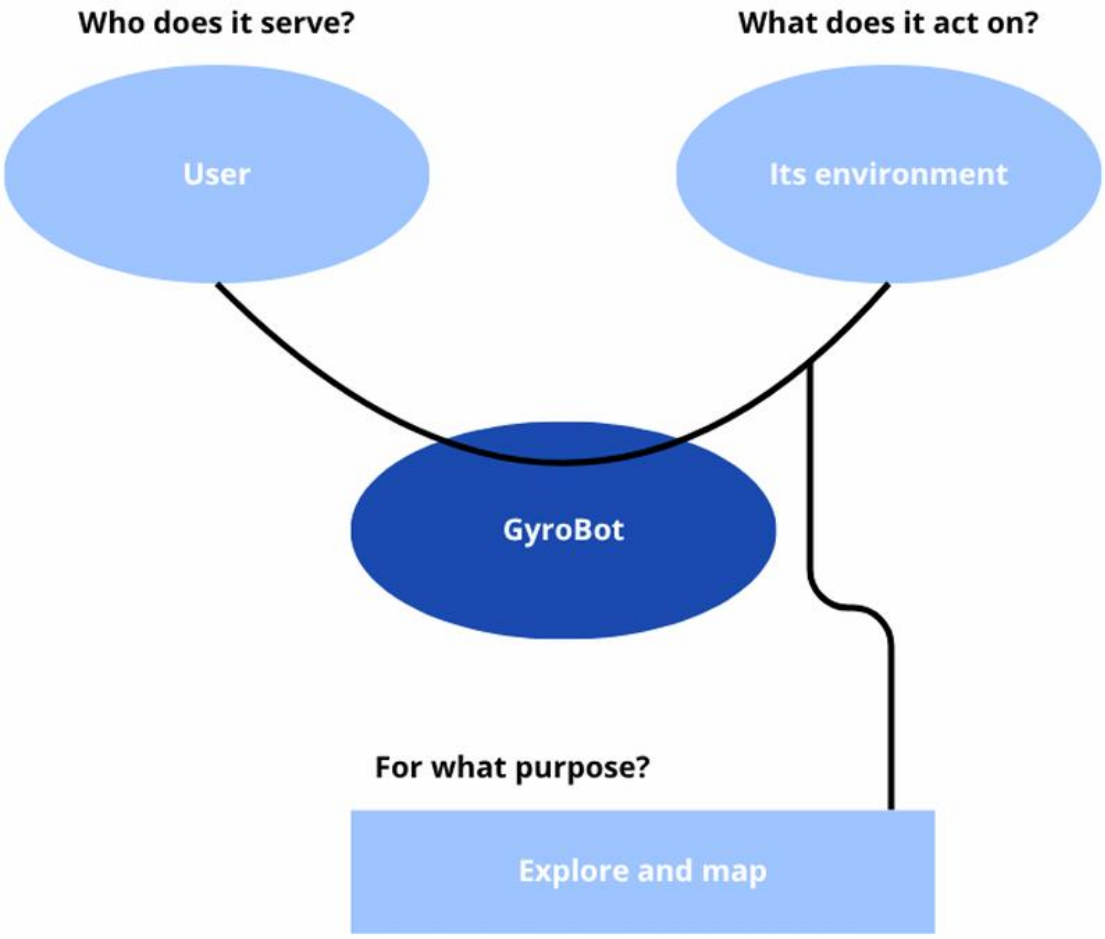



*Figure 4: Horned beast diagram*

The horned beast diagram provides a clear understanding of a user's needs before designing a product or service. In our case, our aim is to map an area and create a map of it, all using the Gyrobot. It will be controlled by a user in various environments, both indoors and outdoors.

### 6.1.2. Octopus diagram

Here are the various block diagrams showing the critical and main functions.

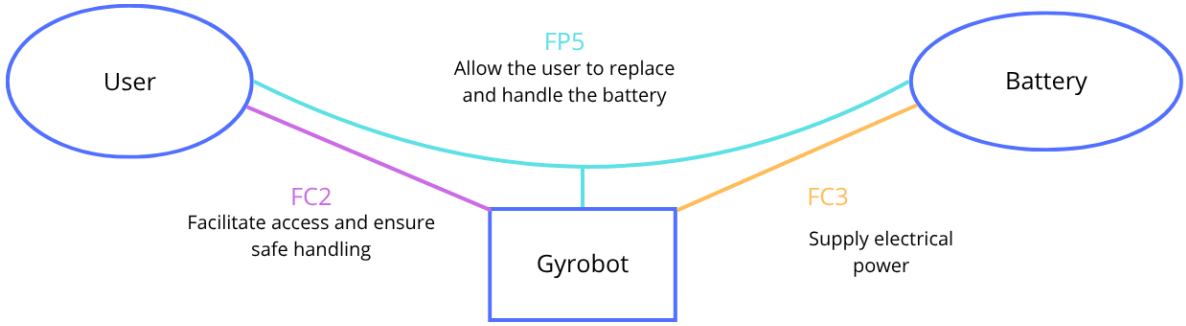



*Figure 5: Energy recharge phase*

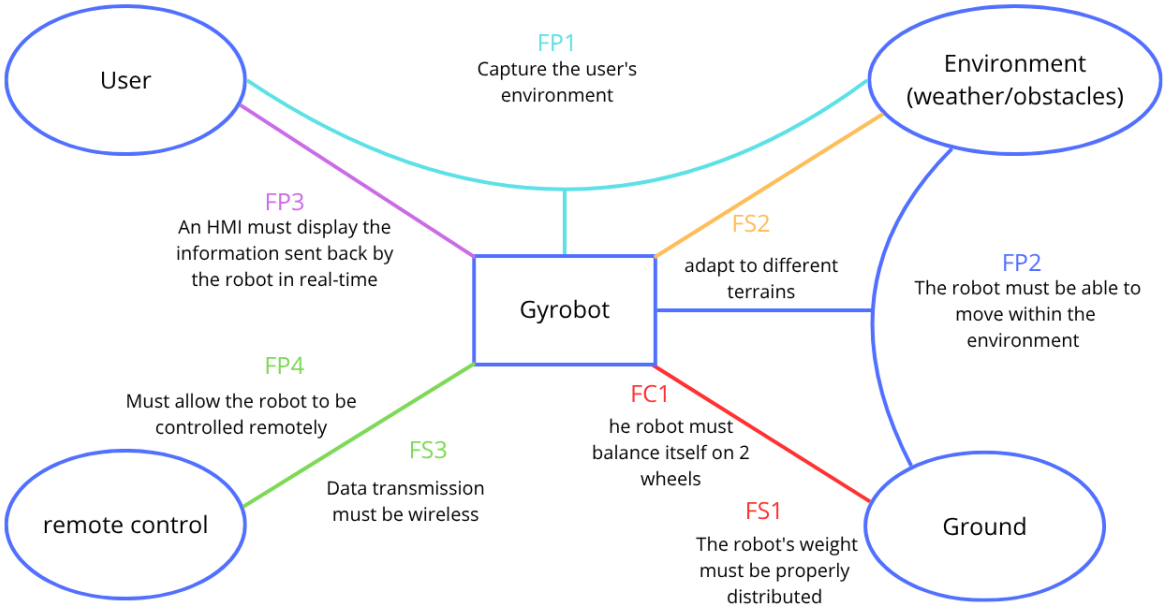



*Figure 6: Usage phase*

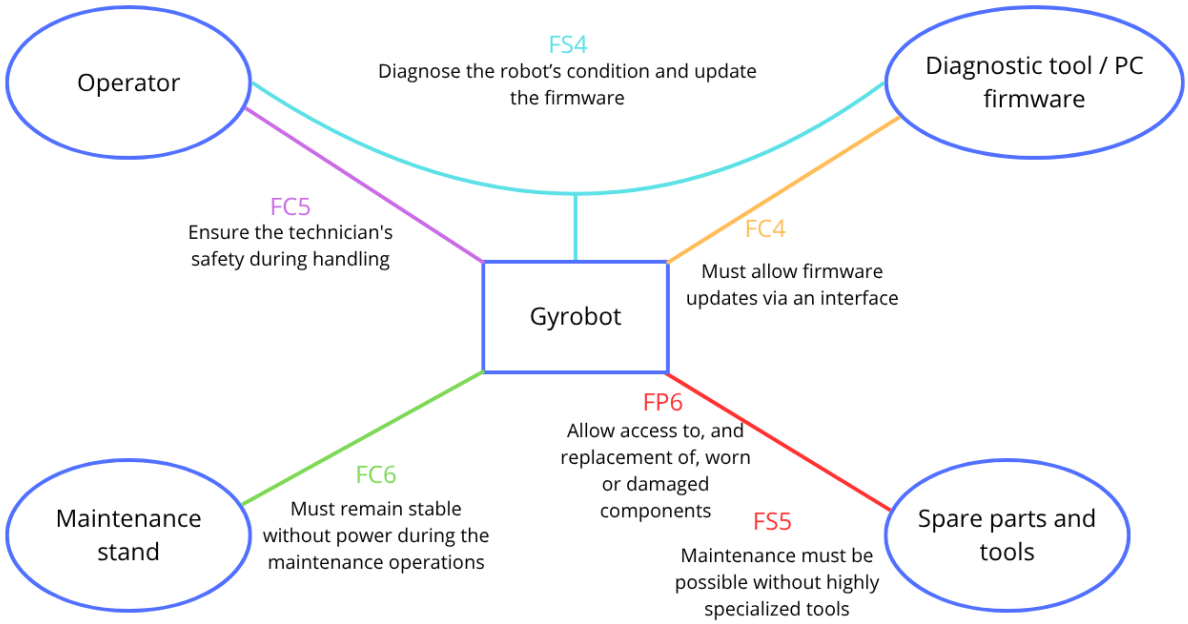



*Figure 7: Maintenance phase*

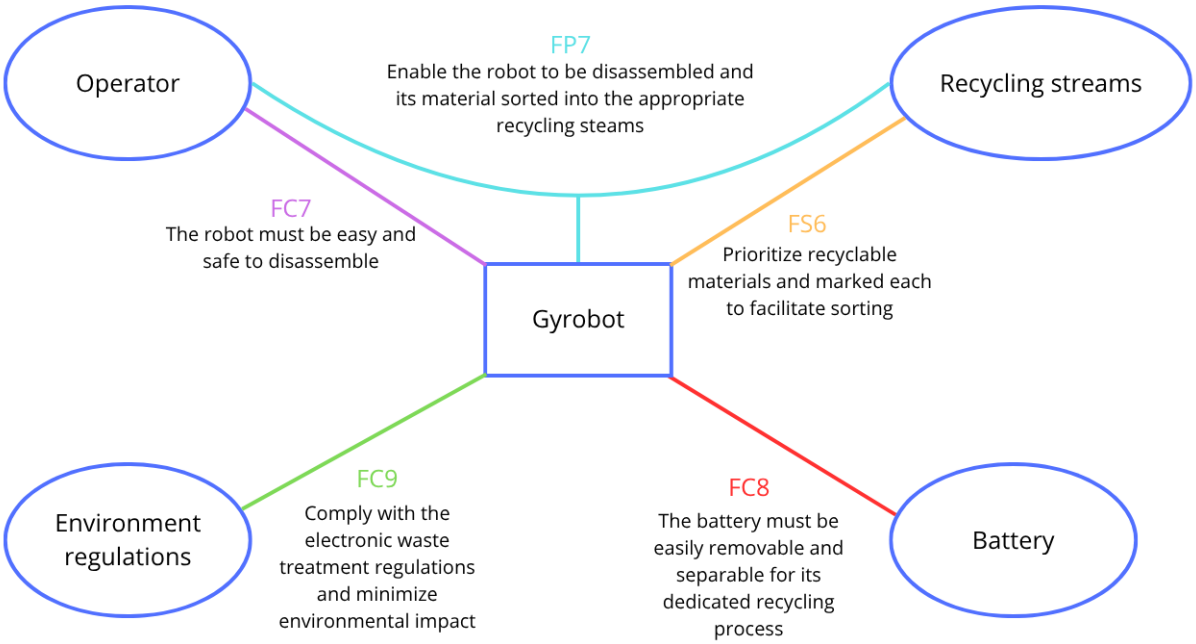



*Figure 8: Recycling phase*

### 6.1.3. FAST diagrams

Using octopus diagrams, we obtain the FAST diagram.




*Figure 9: FAST diagrams*

The FAST diagram enables us to work our way from our requirements to the technical solutions.

## 6.2. Functional table

Using the FAST diagram, we can draw up a functional diagram of the Gyrobot.

| Criteria | Level | Tolerance |
|---|---|---|
| Map accuracy / precision | 5 cm | -3cm + 5cm |
| LiDAR max range | 10m | ± 2m |
| Slip | None | / |
| Tip-over / Fall | None | / |
| Speed | 4 km/h | ± 2 km/h |
| Minimum turning radius (at 4 km/h) | 0.5 m | ± 0.2m |
| Display latency | < 100 ms | / |
| Map refresh rate | 2 Hz | ± 1 Hz |
| Radio control range | > 30 m (interior) | / |
| | > 150 m (exterior) | / |
| Command response time | < 50 ms | / |
| Replacement time | < 1 min | / |
| Replacement time per component | < 1 hour | / |
| Tools required | Standard tools | / |
| Full disassembly time | < 1 hour | |
| Number of separable material families | < 5 | + 2 |
| Maximum climbable slope | > 30° | / |
| Tools required for access | Standard tools | / |
| Connector safety | ? | / |
| Minimum autonomy | > 3min | / |
| Nominal voltage | ? | / |
| Interface type | USB+Wifi | / |
| Supported file format | ? | / |
| Power cut-off before handling | Manual | / |
| Sharp edge | None | / |
| Presence of a dedicated stand | Obligatory | / |
| Stand holding position | Horizontal | / |
| Reclinable | 45° | ± 15° |
| Stand load capacity | > 1,5 kg | / |
| Tools required for disassembly | Standard tools | / |
| Removal time | < 2 min | / |
| Tool-free removal | Yes | / |
| Regulatory compliance | ? | / |
| Hazardous substances presence | None | / |
| Center of Gravity (CoG) position | Centered, above the wheel axis | / |
| Maximum total weight | < 1,5 kg | / |
| Maximum obstacle height | 20 cm | ± 5cm |
| Success rate of landing without falling | 80% | ± 10 % |
| Minimum bandwidth from the robot to remote control | > 30 kbps | / |
| Minimum bandwidth from the robot to the HMI | > 2 Mbps | / |
| Frequency | Wi-Fi 2.4 GHz | / |
| Anomaly detection coverage | 80% | / |
| Firmware update duraction | < 5 min | / |
| Debug communication mode | USB+Wifi | / |
| Tools required for access | Standard tools | / |
| recyclable material by mass | >50 % | / |
| Differents marks on each material | Colored | / |

*Table 2: Functional analysis*

## 6.3. Validations test

To avoid building the robot from scratch only to find that it doesn't work without knowing why, we need to put validation tests in place to test each major component of the robot separately.

Here are the various tests that have been put in place:

- Connection and acknowledgement test
- Command response time test
- Remote control range test
- Communication reliability test
- Data rate and range test to the computer
- Lidar range test
- Map accuracy test
- Display latency and map refresh rate test
- Balance test
- Speed and turning radius test
- Weight test
- Battery replacement test
- Simultaneous operation test

### 6.3.1. Connection and acknowledgement test

We send commands from the remote control and check that the ESP32 receives them and sends back an acknowledgement. The test is successful if the green dot remains lit on the remote control.

### 6.3.2. Testing the response time of the control system

The remote control records the time a command is sent and the time the acknowledgement is received. The difference gives the response time, which must be less than 50 ms. The statistics are displayed in a 'Statistics' tab on the remote control, and the mean and standard deviation are monitored.

### 6.3.3. Remote control range test

Gradually move the remote control away from the robot, first indoors and then outdoors, with and without obstacles. Record the distance once the connection is lost. The test is successful if the connection remains stable for more than 30 metres indoors and more than 150 metres outdoors. Check the green dot and the number of lost packets in the 'Stats' tab on the remote control.

### 6.3.4. Transaction reliability test

Leave the robot to run for 1 minute, whilst moving it around, and note the number of lost packets in the 'Statistics' tab on the remote control, then in the HMI. The test is successful if the loss rate remains below 5 per cent.

### 6.3.5. Testing the data rate and range to the computer

Measure the actual data transfer rate between the Raspberry Pi and the computer. The data transfer rate must remain above 2 Mbps.

### 6.3.6. Lidar range test

An obstacle is placed at increasing distances from the robot, and we check whether it appears on the HMI map. The maximum range should be 10 m ± 2 m. We compare the distance displayed on the HMI with the measured distance to verify reliability at each distance.

### 6.3.7. Map accuracy test

We have a room or object mapped, having first measured its dimensions. We then compare the dimensions shown on the map with those in reality. The test is deemed successful if the deviation falls within the tolerance specified in the specifications (5 cm, -3 / +5 cm).

### 6.3.8. Testing the display latency and refresh rate of the graphics card

The Raspberry Pi stamps each data point with a timestamp upon acquisition, and the computer compares this timestamp with the display time, after synchronising the two. The latency must be less than 100 ms. To calculate the refresh rate, we count the number of map updates over 10 seconds, which should give 2 Hz ± 1.

### 6.3.9. Balance test

The robot is left stationary and then set in motion for several minutes, whilst being given gentle nudges. The robot must neither fall over nor slip. We also check that it can climb up and down a slope steeper than 30°.

### 6.3.10. Speed and turning circle test

A distance of 10 m is marked out and the time taken to cover this distance at maximum speed is recorded: 4 km/h corresponds to approximately 9 s, with a margin of error of ± 2 km/h. To test the cornering, the robot is made to trace a circle, and its radius is measured (0.5 m ± 0.2 m). A visual check is carried out to ensure there is no slippage.

### 6.3.11. Weight test

We weigh the complete robot, including the battery, on a set of scales: it must weigh less than 1.5 kg.

### 6.3.12. Battery replacement test

Someone who did not design the robot replaces the battery, timing the process, without using any tools. The replacement must take less than a minute. A check is then carried out to ensure the robot restarts correctly.

### 6.3.13. Simultaneous testing

We run everything at the same time: balance, movement, controls, Lidar, camera and data transmission to the computer. We monitor the stats tabs for several minutes. The test is successful if the robot does not fall over and if all the previous criteria are met.

# 7. Description of technical choices

## 7.1. Remote control

| 1. Power supply parameters | | |
|---|---|---|
| **Parameter** | **Value** | **Comment** |
| Nominal battery voltage (V) | 7,4 | Battery selected : 2S LiPo (7.4 V nominal) |
| Battery capacity (mAh) | 500 | Battery selected : 2S 500 mAh LiPo |
| Usable capacity fraction | 80% | Assumption : 80 % (to avoid fully discharging the battery) |
| LM2596 output voltage (V) | 5,0 | Set to 5 V to power the ESP32 board |
| Converter efficiency | 80% | Assumption : 85 % (typical 80–90 % for an LM2596) |

| 2. Element power consumption | | | | | |
|---|---|---|---|---|---|
| **Element** | **Qty** | **Unit current (mA)** | **Powered by** | **Total current (mA)** | **Assumption / source** |
| ESP32-S3 (ESP-NOW transmission) | 1 | 120 | 5 V rail | 120,0 | Average during Wi-Fi transmission -> higher peaks |
| OLED screen 0.96" SSD1306 | 1 | 20 | 5 V rail | 20,0 | Screen on and changing image at 2Hz |
| Joystick KY-023 | 2 | 1 | 5 V rail | 2,0 | 2 10 kΩ potentiometers at 3.3 V ≈ 0.7 mA |
| Slide potentiometer | 3 | 0,5 | 5 V rail | 1,5 | 10 kΩ at 3.3 V ≈ 0.33 mA |
| Rotary potentiometer | 1 | 0,5 | 5 V rail | 0,5 | 10 kΩ at 3.3 V ≈ 0.33 mA |
| Signal LED | 1 | 10 | 5 V rail | 10,0 | LED permanently on |
| Digital voltmeter | 1 | 10 | 5V rail | 10,0 | Connected to the output of the 7.4V->5V converter |

| 3. Results | | |
|---|---|---|
| **Indicator** | **Value** | **Comment** |
| Total current on the 5 V rail (mA) | 164,0 | Sum of the elements powered by the converter |
| Power on the 5 V rail (mW) | 820 | P = U * I |
| Current drawn from the battery (mA) | 138,5 | Rail power referred back to the battery (efficiency included) |
| **Estimated battery life (h)** | **2,9** | Usable capacity / current drawn |
| Max. LM2596 current (mA) | 3 000 | Manufacturer value : 3 A |
| Voltage margin Vbattery − Voutput (V) | 2,4 | The LM2596 requires a minimum input difference of about 2 V |
| Sufficient voltage margin ? | YES | To be checked when the battery is discharged |

*Table 3: Remote control, justification for technical choices*

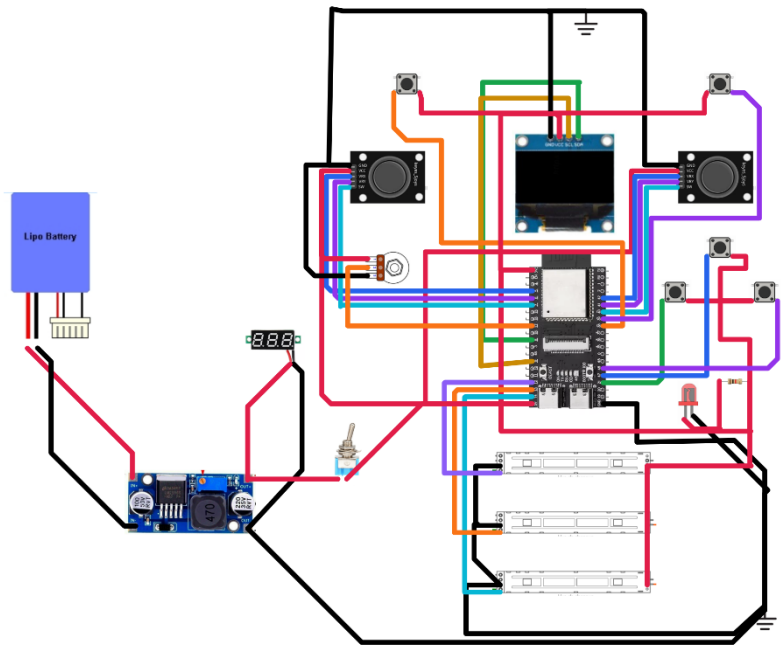



*Figure 10: Electronic schematic for the remote*

## 7.2. Gyrobot

### 7.2.1. Motor selection

**1 Data and formulas**

- Maximum robot mass: M = 1.3 kg
- Gravitational acceleration: g = 9.81 $m/s^2$
- Number of motors: 2 identical motors
- Specifications requirement: Nominal speed v ∈ [2.0; 6.0] km/h (4 ± 2 km/h).
- Fall prevention criterion: Minimum acceleration a_max > 9.81 $m/s^2$ (1.0 g).

**Formulas**

1. Angular speed: ω = N × (2π / 60) (rad/s)
2. Max linear speed: v_max = ω × R (m/s) = 3.6 × (ω × R) (km/h)
3. Total ground thrust (2 motors): F = (2 × τ_stall) / R (N)
4. Max acceleration: a_max = F / M ($m/s^2$) and Margin in g = a_max / 9.81

**Characteristics of the 3 motors**

- **Motor 1 (FIT0520)**: N = 300 RPM (ω = 31.42 rad/s), τ = 3.6 kg·cm = 0.353 N·m (2τ = 0.706 N·m).
- **Motor 2 (FIT0521)**: N = 210 RPM (ω = 21.99 rad/s), τ = 10.0 kg·cm = 0.981 N·m (2τ = 1.962 N·m).
- **Motor 3 (FIT0522)**: N = 100 RPM (ω = 10.47 rad/s), τ = 6.5 kg·cm = 0.637 N·m (2τ = 1.275 N·m).

**2 Maximum speed calculations (v_max)**

| Motor | R = 3.5 cm | R = 4.0 cm | R = 5.0 cm | R = 6.0 cm |
|---|---|---|---|---|
| **FIT0520** (300 RPM) | 3.96 km/h | 4.52 km/h | 5.66 km/h | 6.79 km/h *(Too fast)* |
| **FIT0521** (210 RPM) | 2.77 km/h | 3.17 km/h | 3.96 km/h | 4.75 km/h |
| **FIT0522** (100 RPM) | 1.32 km/h | 1.51 km/h | 1.88 km/h | 2.26 km/h |

*Table 4: Maximum linear speed as a function of wheel radius (allowed range: 2 to 6 km/h).*

**Speed conclusion:**

- **FIT0522 (100 RPM)**: Eliminated. Far too slow for wheel radii from 3.5 to 5 cm (< 2.0 km/h, non-compliant with specifications).
- **FIT0520 (300 RPM)**: Compliant up to 5 cm, but exceeds the maximum speed limit at 6 cm (6.79 km/h > 6.0 km/h).
- **FIT0521 (210 RPM)**: Compliant. Strictly within the required [2.0 - 6.0] km/h range for all evaluated radius.

**3 Maximum Acceleration Calculations (a_max for M = 1.3 kg)**

*Total thrust is $F = (2 \times \tau) / R$ and acceleration is $a_{max} = F / 1.3$*

| **Motor** | **R = 3.5 cm** | **R = 4.0 cm** | **R = 5.0 cm** | **R = 6.0 cm** |
|---|---|---|---|---|
| **FIT0520** (3.6 kg·cm) | 15.5 m/s$^2$ (1.58 g) | 13.6 m/s$^2$ (1.38 g) | 10.9 m/s$^2$ (1.11 g) | 9.0 m/s$^2$ (0.92 g) *(Fall)* |
| **FIT0521** (10.0 kg·cm) | 43.1 m/s$^2$ (4.40 g) | 7.7 m/s$^2$ (3.85 g) | 30.2 m/s$^2$ (3.08 g) | 25.2 m/s$^2$ (2.56 g) |
| **FIT0522** (6.5 kg·cm) | 28.0 m/s$^2$ (2.86 g) | 24.5 m/s$^2$ (2.50 g) | 19.6 m/s$^2$ (2.00 g) | 16.4 m/s$^2$ (1.67 g) |

*Table 5: Maximum available acceleration to prevent tipping for a 1.3 kg robot.*

**Acceleration conclusion:**

- To stabilize the inverted pendulum, the acceleration must strictly exceed gravity ($a > 1.0\ g = 9.81\ m/s^2$).
- **FIT0520 (3.6 kg·cm)**: As soon as the radius reaches 6 cm, acceleration drops to 0.92 g (< 1.0 g): the robot cannot recover and tips over. Even at 4 cm (1.38 g), the margin is very narrow.
- **FIT0521 (10.0 kg·cm)**: Delivers between 2.56 g and 4.40 g of acceleration, guaranteeing that the robot will never stall or tip over.

**4 Selected Choice**

1. **FIT0522 (100 RPM; 6.5 kg·cm)**: Rejected because it is far too slow.
2. **FIT0520 (300 RPM; 3.6 kg·cm)**: Rejected due to insufficient torque and risk of balancing failure at 1.3 kg.

3. **FIT0521 (210 RPM; 10.0 kg·cm)**: Selected. The only motor that simultaneously satisfies the speed requirement and provides ample acceleration margin across all radii.

4. **Optimal Wheel Radius**: R = 3.5 cm to 4.5 cm (diameter of 70 to 90 mm), providing ≈ 3.2 km/h and nearly 4 g of acceleration.

### 7.2.3. Electronic schematic

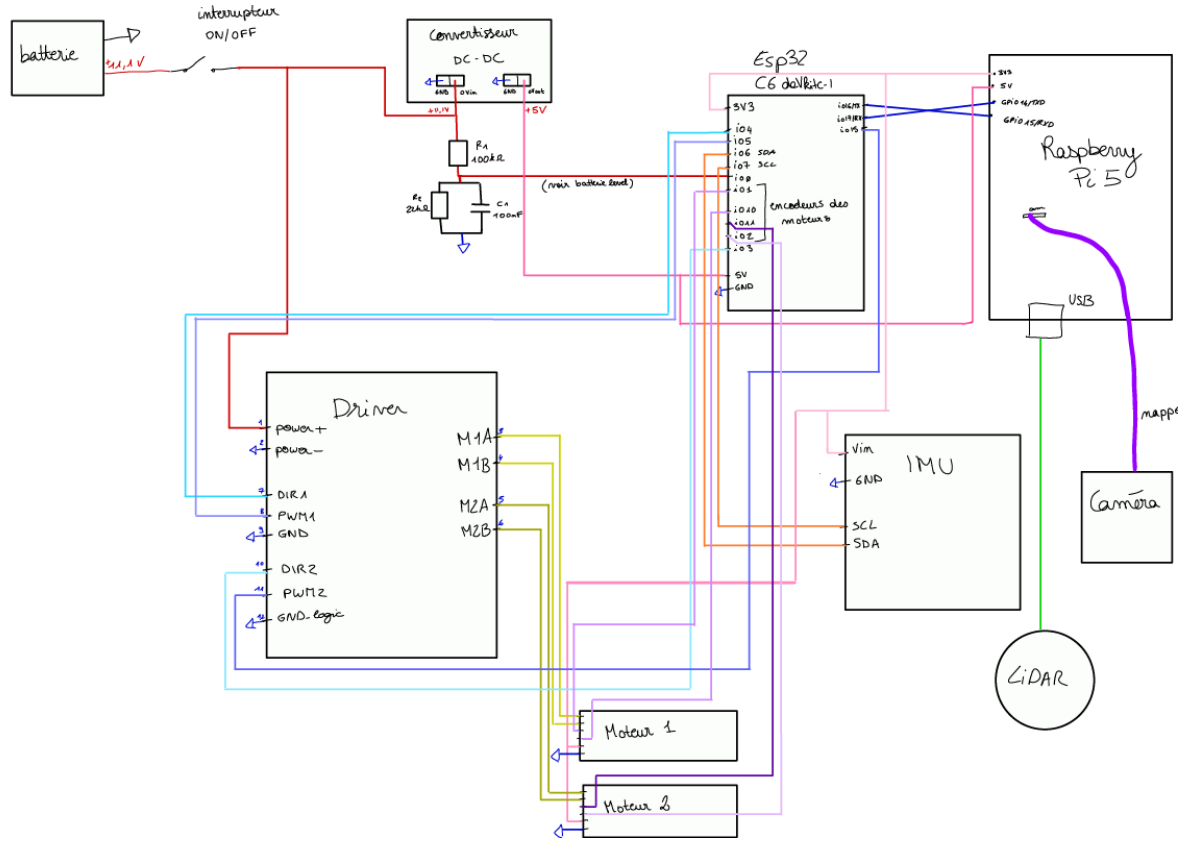



*Figure 11: electronic schematic for the robot*

# 8. RSE

The Gyrobot project integrates an eco-design approach to limit our carbon footprint. Instead of making parts far away, all the structure is manufactured locally by 3D printing using PLA, a bio-based and recyclable plastic. To make the sorting easier at the end of life (FS6), recycling symbols are modelled directly in relief on the CAD parts. To go further in accessibility, each material or sorting instruction has its own geometric logo in relief and a colour code. These colours were specifically chosen to be easily seen by colourblind people (yellow, dark blue, white, black, magenta).

For the electronic part, designing and routing a custom motherboard allows us to connect the microcontrollers and components directly through copper tracks. This avoids using dozens of flying wires. It reduces the number of plastic cables used and makes the system much tougher.

The robot was also designed to last in time and to fight against obsolescence (meaning avoiding throwing it away at the first broken component). All the architecture allows us to easily access fragile parts to change them quickly (FP6). The assembly is done using only standard screws, so the structure can be disassembled very simply (FC7) with basic screwdrivers without special tools (FS5). The battery, for example, can be removed very easily to be sent to its specific recycling channel which makes recycling much easier (FC8).

This design choice allows us to respect the European WEEE directive (Waste Electrical and Electronic Equipment) which imposes strict rules to limit the pollution of this kind of system (FC9).

Finally, beyond the environmental impact, the Gyrobot has a real social utility focused on safety at work. The main goal is to map inaccessible, unstable or dangerous places instead of humans. Whether it is to explore a building for a rescue or to inspect unreachable industrial environments, all the necessary spatial data is collected by the robot without ever putting the lives of rescue teams or technicians in danger.

# FMEA Table

FMEA is an analysis table that helps us figure out what could go wrong when our robot is exploring. Since the robot only uses two wheels, keeping its balance and staying connected to the remote are extremely important. This table lists the main problems the system might face. We give each risk a score based on how bad it is (Severity), how often it happens (Frequency) and how hard it is to spot (Detection). This will help us to focus on the biggest dangers so we can figure out the best ways to prevent them.

| **Risks** | **Potential Effects** | **Probable Causes** | **G** | **F** | **D** | **C** | **Preventive Actions** |
|---|---|---|---|---|---|---|---|
| *Computer overload or balancing system failure* | The robot stops balancing and crashes, breaking the Lidar and camera. | The robot's microcontroller gets too busy processing maps and images causing a small freeze that stops the balancing system. | 4 | 3 | 4 | 48 | Use two separate microcontrollers: a small one just to keep the robot balanced and another one for the map and camera. |
| *Wheels slipping* | The robot falls right away. If the wheels slide, the robot cannot balance itself and will get damaged. | Driving on mud, wet grass or loose dirt. The two wheels need a good grip on the ground to keep the robot upright. | 4 | 3 | 2 | 24 | Use tires with strong grip. Add physical stands that drop down to catch the robot if it starts to slip. |
| *Falling downstairs or holes* | The robot breaks completely, destroying the expensive sensors. | The Lidar only looks straight ahead and cannot see holes or stairs. The robot drives over the edge. | 4 | 2 | 2 | 16 | Rely on the real-time camera video feed on the HMI so the pilot can visually detect stairs and drop-offs. Mount the camera with a slight angle to |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | maximize ground visibility. |
| *Losing connection to the remote* | The user cannot control the robot anymore, and the computer stops receiving the map. | The robot is too far away or thick walls block the signal. | 3 | 2 | 2 | 12 | Make the robot stop moving automatically if it loses the signal. Show the connection strength on the screen. |
| *Battery dying too fast* | The robot turns off and gets stuck in an area you cannot reach before the 40 minutes are over. | Driving on tough surfaces like thick grass or mud uses too much energy. | 3 | 2 | 2 | 12 | Show the battery level on the remote and make it flash red when it is almost empty. |
| *Map system not working* | The robot cannot create the map or show it on the computer screen. | Something is blocking the Lidar sensor, or the computer software crashes. | 2 | 1 | 2 | 4 | Put a protective cover over the sensors and test the computer software many times. |

*Table 6: FMEA table*

**Legend:**

- *G (Severity)*: 1 to 4
- *F (Frequency)*: 1 (Rare) to 4 (Frequent)
- *D (non-detection)*: 1 (Immediately detectable) to 4 (Hard to predict)
- *C (Criticality)* = G × F × D

# Conclusion

To conclude, this project is going to be very demanding, particularly in terms of time. Nevertheless, our team has shown great commitment, which is spurring us on to persevere and achieve our goal of producing a prototype before the end of the term, featuring the following key functions:

- To balance,
- Mapping using lidar,
- Move around using a remote control.

# Appendices

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